

FOLIO

Personal Finance App

Product Requirements Document

Version 1.0 · MVP Specification

June 2026

1. Product Overview

Folio is a mobile-first personal finance app built for a single user to track income, expenses, savings, and outstanding loans — all denominated in GHS — with month-to-month history.

The app answers one question every time it is opened:

Core Question

Am I financially safe for the rest of this month, and who owes me money?

1.1 Design Philosophy

- No spreadsheet-like tables on the home screen
- Every key metric visible without drilling into sub-screens
- Business logic enforced by the app, not by user discipline
- Loan repayments are never treated as income
- Savings withdrawals carry a persistent accountability cost

1.2 Out of Scope for V1

Feature	Reason
Charts / Analytics	V2
PDF / Excel Export	V2
Push Notifications	V2
Recurring Transactions	V2
Multi-currency	Not needed — GHS only
Authentication / Login	Single-user personal app
Web dashboard	Mobile-only
AI categorisation	V2
Bank integrations	V2

2. Technical Stack

Layer	Decision
Mobile Framework	React Native + Expo
Language	TypeScript

State Management	Zustand
Local Persistence	expo-sqlite (offline-first cache)
Backend	Next.js API routes (existing infra)
Database	MongoDB (existing infra)
Sync Strategy	Local SQLite as write-through cache; sync to MongoDB on connection
Currency	GHS hardcoded — no multi-currency
Auth	None for V1 — single user

Sync Rationale

SQLite ensures the app works offline. MongoDB provides cloud backup so a lost or replaced phone does not destroy financial history. Sync happens on every successful write when connected, and batches on reconnect when offline.

3. Data Model

3.1 Accounts

An account represents a place where money is stored.

```
interface Account {
  id: string // UUID
  name: string // e.g. "Salary Account", "MoMo", "Cash"
  type: 'checking' | 'savings' | 'cash' | 'momo'
  isProtected: boolean // true = savings; excluded from Safe-to-Spend
  balance: number // derived from transactions, cached for performance
  createdAt: string // ISO 8601
  updatedAt: string
}
```

Protected Accounts

Any account where `isProtected = true` is treated as savings. Its balance does not contribute to Safe-to-Spend Today. Moving money OUT of a protected account increases Discipline Debt.

3.2 Transactions

```
interface Transaction {
  id: string
  type: 'income' | 'expense' | 'transfer'
  amount: number
  date: string
  categoryId: string | null // required for expense
}
```

```

    fromAccountId: string | null    // required for expense and transfer
    toAccountId:   string | null    // required for transfer
    note:          string | null
    reason:        string | null    // required for protected → spendable transfer
    createdAt:     string
    updatedAt:     string
  }

```

3.3 Categories

```

interface Category {
  id:         string
  name:       string
  type:       'income' | 'expense'
  createdAt:  string
  updatedAt:  string
}

```

3.4 Budget

Planned spending per category, per month. Actuals are always derived from transactions — never entered manually.

```

interface Budget {
  id:           string
  categoryId:   string
  month:        string    // format: YYYY-MM
  plannedAmount: number
  createdAt:    string
  updatedAt:    string
}

```

3.5 Month

```

interface Month {
  id:           string    // format: YYYY-MM
  startDate:    string    // ISO 8601
  createdAt:    string
}

```

Month is a boundary record only. No calculated totals stored here — all figures are derived from transactions.

3.6 Loans

Tracks money lent to other people. Loan events do not touch income or expense categories.

```

interface Loan {
  id:           string
  borrowerName: string
}

```

```

    principal:    number    // original amount lent (can be cumulative if ongoing
tab)
    dateLent:      string
    expectedRepaymentDate: string | null    // optional; triggers overdue warning on
home screen
    note:          string | null
    sourceAccountId: string    // account debited when loan was made
    createdAt:     string
    updatedAt:     string
}

interface LoanPayment {
    id:            string
    loanId:        string
    amount:        number
    date:          string
    destinationAccountId: string    // account credited on repayment
    note:          string | null
    createdAt:     string
}

```

Key Rule

Loan disbursements debit an account and create a Loan record. Repayments credit an account and reduce the loan outstanding balance. Neither event appears in income or expense categories.

3.7 Discipline State

Tracks how much has been withdrawn from protected savings without being repaid through extra savings contributions. Persists across months and never resets automatically.

```

interface DisciplineState {
    id:            string
    totalWithdrawnFromSavings: number
    totalExtraSavings: number
    createdAt:     string
    updatedAt:     string
}

// Discipline Debt (derived, never stored)
disciplineDebt = Math.max(0, totalWithdrawnFromSavings - totalExtraSavings)

```

4. Business Rules

4.1 Safe-to-Spend Today

The primary metric. Shown as the dominant figure on the home screen.

```

usableBalance    = sum of balance across all accounts where isProtected = false
daysRemaining   = days left in current month including today
safeToSpendToday = usableBalance / daysRemaining

```

Value	UI State
Positive (comfortable)	Green
> 0 but below low threshold (configurable, default GHS 50)	Yellow — "Spending carefully"
Negative	Red — "You are financially off-track for this month"

Recalculation Trigger

Safe-to-Spend must recalculate after every transaction and automatically on every new day (daysRemaining changes at midnight).

4.2 Income Rule — Mandatory Savings

Every income transaction must split into a spendable portion and a protected savings portion.

- Minimum savings = 10% of income received
- User may save more than 10%, never less
- App pre-fills savings field at 10% and blocks values below it
- If savings > 10%, the excess reduces Discipline Debt

```

minimumSavings = incomeAmount * 0.10
extraSavings    = selectedSavings - minimumSavings

// Effects on save:
protectedAccount.balance += selectedSavings
spendableAccount.balance += (incomeAmount - selectedSavings)
if (extraSavings > 0) disciplineState.totalExtraSavings += extraSavings

```

4.3 Expense Rule

- Must debit a spendable account (never a protected account)
- Must be linked to a category
- Recalculates Safe-to-Spend immediately on save
- If Safe-to-Spend goes negative after save, show warning — but do not block the transaction

4.4 Transfer Rules

Direction	Safe-to-Spend Effect	Discipline Debt Effect
Spendable → Spendable	No change	No change
Spendable → Protected	Decreases	Can decrease if flagged as debt repayment

Protected → Spendable	Increases	Increases — requires reason + confirmation modal
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Protected → Spendable Warning Copy

You are withdrawing from protected savings. This will increase your Discipline Debt and should only be done intentionally.

4.5 Loan Rules

- Disbursing a loan debits a spendable account and creates a Loan record
- A loan does not appear as an expense
- Receiving a repayment credits a spendable (or protected) account and reduces loan outstanding balance
- A repayment does not appear as income
- Outstanding balance = principal - sum of all LoanPayments for that loan
- Partial repayments are supported — a borrower may pay back in multiple instalments
- A borrower (e.g. Bridget, Mina) may have multiple active loans — the app tracks each separately and shows total owed per borrower

4.6 Budget Rules

- Planned amounts are set per category per month
- Actual amounts are always derived from transactions — never entered manually
- A category with no Budget record for the current month shows planned = GHS 0.00
- Budget carries no meaning for income categories — it applies to expenses only

5. Screens & UI

5.1 Navigation

Bottom tab navigation with five tabs:

1. Home
2. Transactions
3. Accounts
4. Loans
5. Settings

A floating "+ Add" button appears on Home and Transactions screens for quick transaction entry.

5.2 Home Screen

The most important screen. No tables, no charts. Three primary metrics + one summary card.

Element	Detail
Safe-to-Spend Today	Large card, top of screen. Color-coded: green / yellow / red. Updates in real time.
Days Remaining	Smaller card below. Plain count of days left in month.
Discipline Debt	Shown only when > 0. Card with red accent. Tapping opens explanation modal.
Loans Summary Card	"You are owed GHS X across N borrowers." Turns yellow with overdue count when any loan has passed its expectedRepaymentDate. Taps through to Loans screen.
Quick Add Button	Floating action button. Opens Add Transaction sheet.

5.3 Add Transaction Flow

Segmented control at top: Expense | Income | Transfer. Fields change based on selection.

Expense Fields

- Amount
- Account (spendable accounts only)
- Category
- Date (defaults to today)
- Note (optional)

If Safe-to-Spend becomes negative on save: show inline warning before confirming.

Income Fields

- Amount
- Spendable account to receive usable portion
- Protected account for savings
- Savings amount (pre-filled at 10%, editable upward only)
- Date (defaults to today)
- Note (optional)

Live preview beneath savings field:

Income received:	GHS 4,000	
Minimum savings:	GHS 400	
Your savings:	GHS 600	← editable
Spendable amount:	GHS 3,400	← auto-calculated

Transfer Fields

- From account
- To account
- Amount
- Date (defaults to today)
- Note (optional)
- Reason (required only when From = protected, To = spendable)

For Spendable → Protected transfers: if Discipline Debt exists, show toggle: "Count this as savings debt repayment?"

5.4 Transactions Screen

Chronological list of all transactions. Each row shows:

- Type icon (income / expense / transfer / loan out / loan repayment)
- Description / note
- Category (if expense)
- Account
- Amount with sign (green positive, red negative)
- Date

Filterable by: month, account, category, type. Tapping a row shows full detail.

5.5 Accounts Screen

List of all accounts. Each card shows:

- Account name and type
- Current balance (derived)
- Protected badge if isProtected = true

Actions: Create account, Edit name/type. Delete is disabled if transactions exist against the account.

5.6 Loans Screen

Two sections: Active Loans and Settled Loans.

Active Loans List

- Borrower name
- Original principal
- Amount repaid to date
- Outstanding balance (prominent)
- Expected repayment date (if set) — shown in red with "Overdue" badge when today is past that date

- Date of last activity

Tapping a borrower opens their loan detail screen showing all disbursements and repayments in a timeline.

Loan Detail Screen

- Record a repayment (amount, date, destination account, optional note)
- Add a new loan to the same borrower (creates a new Loan record)
- Edit expected repayment date
- Mark loan as settled (moves to Settled section)

5.7 Settings Screen

Setting	Detail
Minimum savings rate	Displays 10%. Locked in V1.
Safe-to-Spend low threshold	Default GHS 50/day. Editable. Controls yellow warning trigger.
Categories	Add, rename, or archive expense and income categories.
Default categories	Pre-seeded on first launch (see Section 6).

6. Seed Data (First Launch)

6.1 Default Accounts

Account Name	Type / Protected
Salary Account	checking / spendable
Mobile Money	momo / spendable
Cash	cash / spendable
Savings Account	savings / protected

6.2 Default Categories

Expense Categories	Income Categories
Bills	Salary
Communications	Side Gigs
Education	Deposits

Entertainment	Bonus
Food	
Health	
Insurance	
Miscellaneous	
Transport	
Work	

6.3 Other Seed Records

- One DisciplineState row with all values at 0
- One Month record for the current YYYY-MM

7. Core Service Functions

Business logic must live in service files, not in UI components. UI calls clean functions only.

7.1 addIncome

```
addIncome({
  amount,
  spendableAccountId,
  protectedAccountId,
  selectedSavingsAmount, // must be >= amount * 0.10
  categoryId,
  date,
  note,
})
```

7.2 addExpense

```
addExpense({
  amount,
  accountId, // must be spendable
  categoryId,
  date,
  note,
})
```

7.3 addTransfer

```

addTransfer({
  amount,
  fromAccountId,
  toAccountId,
  date,
  note,
  reason,                // required if protected → spendable
  countsAsDebtRepayment, // boolean, only relevant for spendable → protected
})

```

7.4 addLoan

```

addLoan({
  borrowerName,
  amount,
  sourceAccountId,
  date,
  expectedRepaymentDate, // optional
  note,
})

```

7.5 recordLoanRepayment

```

recordLoanRepayment({
  loanId,
  amount,
  destinationAccountId,
  date,
  note,
})

```

7.6 Calculation Helpers

```

calculateSafeToSpendToday(accounts, currentDate): {
  usableBalance: number,
  daysRemaining: number,
  safeToSpendToday: number,
}

calculateDisciplineDebt(state: DisciplineState): number
  → Math.max(0, state.totalWithdrawnFromSavings - state.totalExtraSavings)

getLoanOutstanding(loanId, payments): number
  → loan.principal - sum(payments where loanId matches)

getMonthSummary(month: string): {
  totalIncome: number,
  totalExpenses: number,
  netSavings: number, // endBalance - startBalance of spendable accounts
  plannedVsActual: { categoryId, planned, actual }[]
}

```

8. Validation Rules

Rule	Detail
All amounts	Must be > 0
Income savings	Must be $\geq \text{income} * 0.10$
Expense account	Must be spendable (isProtected = false)
Transfer accounts	From and To cannot be the same account
Protected → Spendable transfer	Reason field must not be empty
Loan disbursement account	Must be spendable
Loan repayment amount	Cannot exceed outstanding balance
Expected repayment date	Must be after dateLent if provided. Optional — app does not block saving without it.
Category on expense	Required — cannot save without selecting one
Date fields	Cannot be in the future (except for scheduled — V2)

9. MVP Completion Criteria

The app is shippable when all of these pass:

1. User can create spendable and protected accounts
2. User can add income; app enforces minimum 10% savings split
3. User cannot save less than 10% of received income
4. User can save more than 10%; extra portion reduces Discipline Debt
5. User can add expenses from spendable accounts only
6. Expenses update Safe-to-Spend Today immediately
7. User can record transfers between accounts
8. Spendable → spendable transfers are balance-neutral on Safe-to-Spend
9. Spendable → protected transfers reduce Safe-to-Spend Today
10. Protected → spendable transfers require a reason and confirmation modal
11. Protected → spendable transfers increase Discipline Debt
12. User can record a loan disbursement; balance of source account decreases
13. Loan disbursement does not appear as an expense
14. User can record a loan repayment; loan outstanding balance decreases
15. Loan repayment does not appear as income
16. Partial repayments are supported

17. Loans can have an optional expected repayment date
18. Loans past their expected repayment date show an "Overdue" badge on the Loans screen
19. Home screen loans card turns yellow and shows overdue count when any loan is past its due date
20. Safe-to-Spend Today appears prominently on Home, color-coded
21. Discipline Debt appears on Home when > 0
22. Loans summary card on Home shows total owed across all borrowers
23. Days remaining in month appears on Home
24. Budget: user can set planned amounts per category per month
25. Budget: actual vs planned shown per category (derived from transactions)
26. Month-to-month history is preserved and browsable
27. All data syncs to MongoDB when online; app works offline via SQLite
28. All amounts display in GHS

10. UX Copy Reference

Situation	Copy
Safe-to-Spend negative	"You are financially off-track for this month."
Safe-to-Spend yellow zone	"Spending carefully."
Income below savings minimum	"Minimum savings is 10% of received income."
Savings withdrawal warning	"You are withdrawing from protected savings. This will increase your Discipline Debt and should only be done intentionally."
Discipline Debt label	"Discipline Debt"
Discipline Debt description	"Money you have withdrawn from savings that has not yet been repaid through extra savings contributions."
Expense would make Safe-to-Spend negative	"This expense will put you off-track for the rest of the month. Continue?"
Loans summary (home)	"You are owed GHS X across N borrowers."
Loans summary — overdue loans exist	"You are owed GHS X across N borrowers. N overdue."
Loan overdue badge (Loans screen)	"Overdue"
No active loans	"No outstanding loans."
Loan fully repaid	"Loan settled."